Gabor Vector Fields: Kernel Smoother Model Comparison

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May 22, 2026

1 Model

Each Gabor observation is a coordinate $z_i = (x_i, y_i)$ and a local pitch angle α'_i. The global line-field angle used for fitting is

$$\phi_i = \operatorname{atan2}(y_i, x_i) + \alpha'_i.$$

Because the data are axial line fields, the smoother is applied to the doubled-angle embedding

$$u_i = (\cos 2\phi_i, \sin 2\phi_i).$$

For a target point z, the Nadaraya–Watson kernel smoother is

$$\hat{u}(z) = \frac{\sum_i K(z, z_i) u_i}{\sum_i K(z, z_i)}, \qquad \hat{\phi}(z) = \frac{1}{2} \operatorname{atan2}(\hat{u}_2(z), \hat{u}_1(z)).$$

The RBF kernel is

$$K_{\mathrm{RBF}}(z, z'; h) = \exp\left[-\frac{\|z - z'\|^2}{2h^2}\right],$$

and the multiplicative RBF–von-Mises kernel is

$$K_{\mathrm{mult}}(z, z'; h, \kappa) = \exp\left[-\frac{\|z - z'\|^2}{2h^2}\right] \exp\left[\kappa \cos\{\vartheta(z) - \vartheta(z')\}\right], \quad \vartheta(z) = \operatorname{atan2}(y, x).$$

2 Hyperparameter Selection

The sweep is fixed across all datasets:

$$h \in \operatorname{geomspace}(1, 1500, 72),$$

and for the multiplicative kernel

$$\kappa \in \{0, 0.125, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 64\}.$$

Each candidate is scored by shuffled 10-fold held-out axial reconstruction RSS using random seed 42. This is used only as an anti-overfitting check for selecting the smoothness of the streamline description.

3 Results

Model	Mean RSS	Median RSS	Mean MAE deg	h med	κ med
RBF-von-Mises kernel smoother	0.3577	0.3496	25.5983	29.9372	0.75
RBF kernel smoother	0.3608	0.354	25.6744	29.9372	

Table 1: Selected kernel smoother performance across the Gabor vector-field exports.

Model	Mean RSS	Median RSS	Mean MAE deg
RBF-von-Mises kernel smoother	0.3577	0.3496	25.5983
RBF kernel smoother	0.3608	0.354	25.6744
Parametric continuous p	0.5588	0.5885	35.2181
Parametric p=1 (Archimedean)	0.5949	0.5892	36.668
Parametric p=0 (Logarithmic)	0.6362	0.6331	38.2387
Parametric p=2 (Fermat)	0.6567	0.6875	38.8822

Table 2: Held-out axial reconstruction comparison including the parametric spiral families.

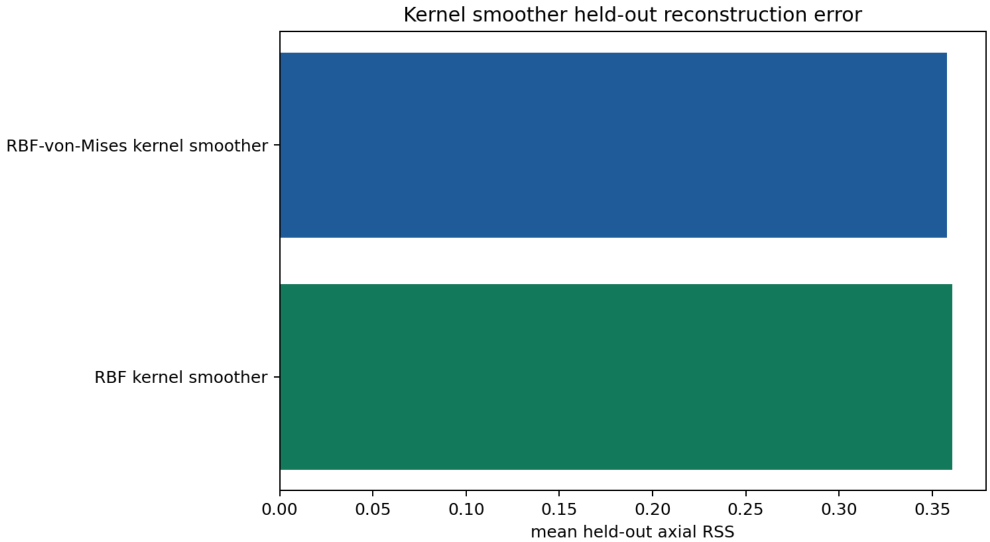

Figure 1: Mean held-out axial RSS for the two kernel smoothers.

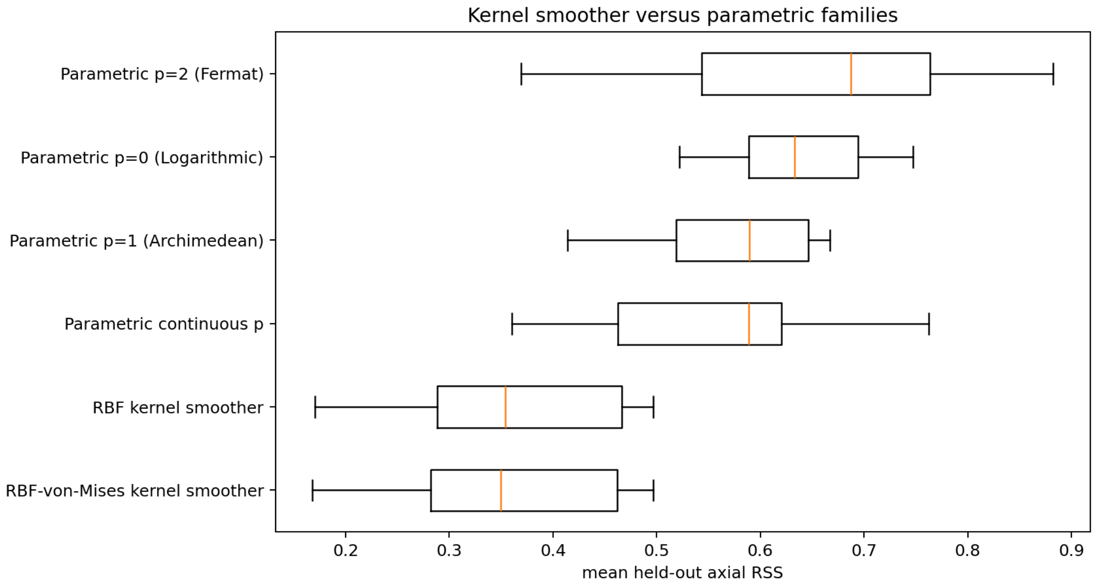

Figure 2: Held-out axial RSS distribution comparing selected kernel smoothers with the parametric spiral families.

4 Streamline Overlays

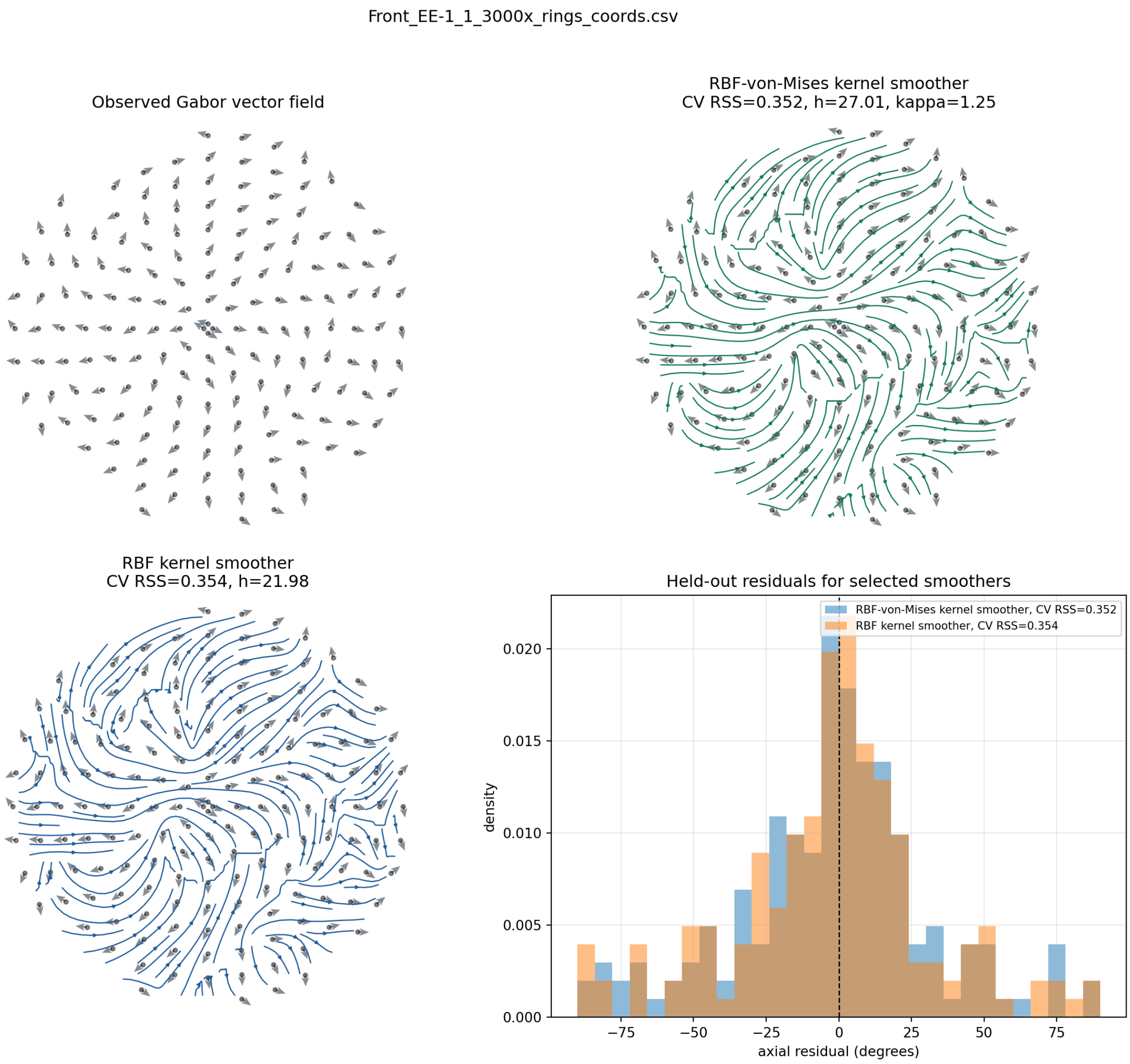

Figure 3: Kernel-smoother streamline overlays for `Front_EE-1_1_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.

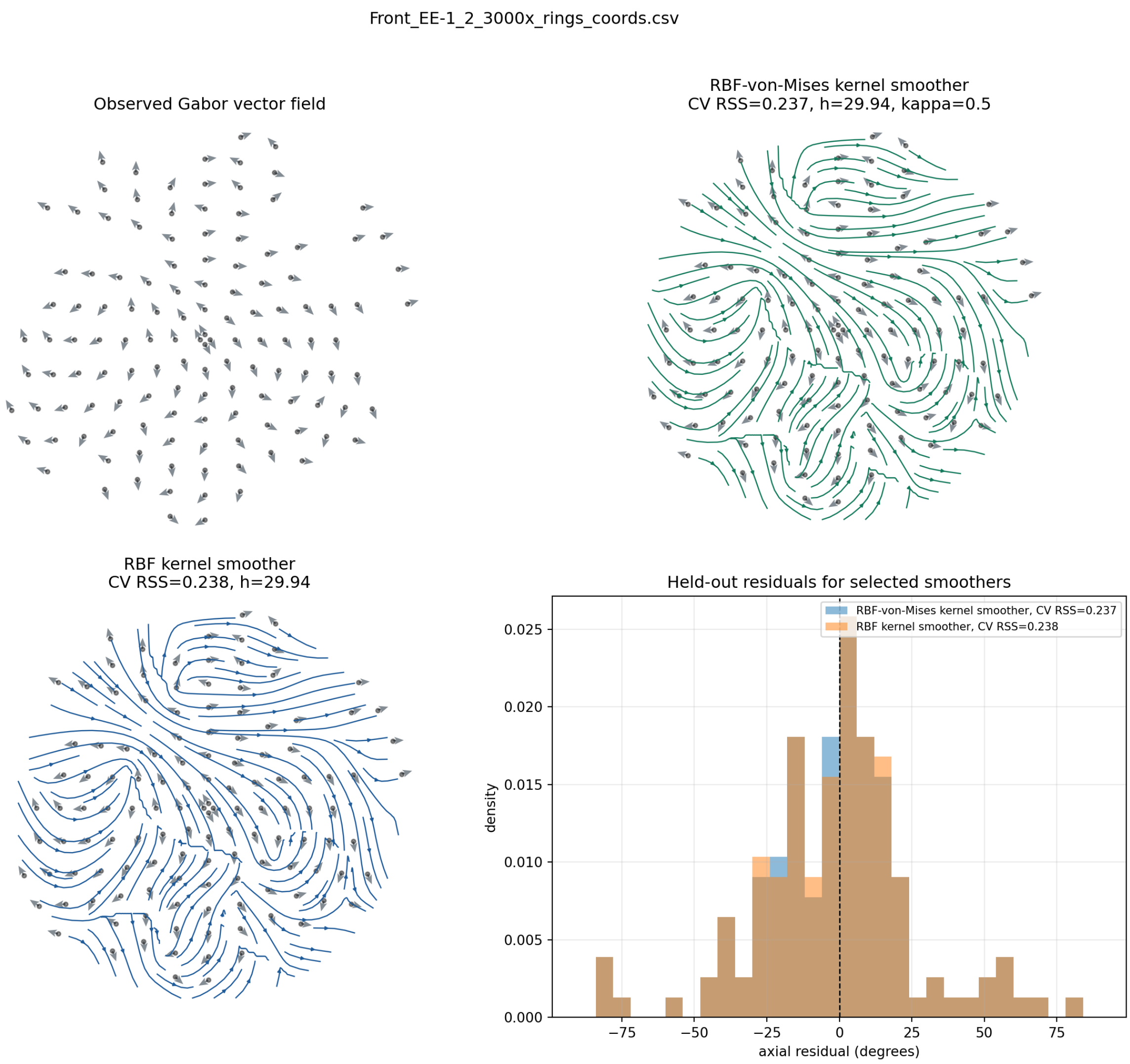

Figure 4: Kernel-smoother streamline overlays for `Front_EE-1_2_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.

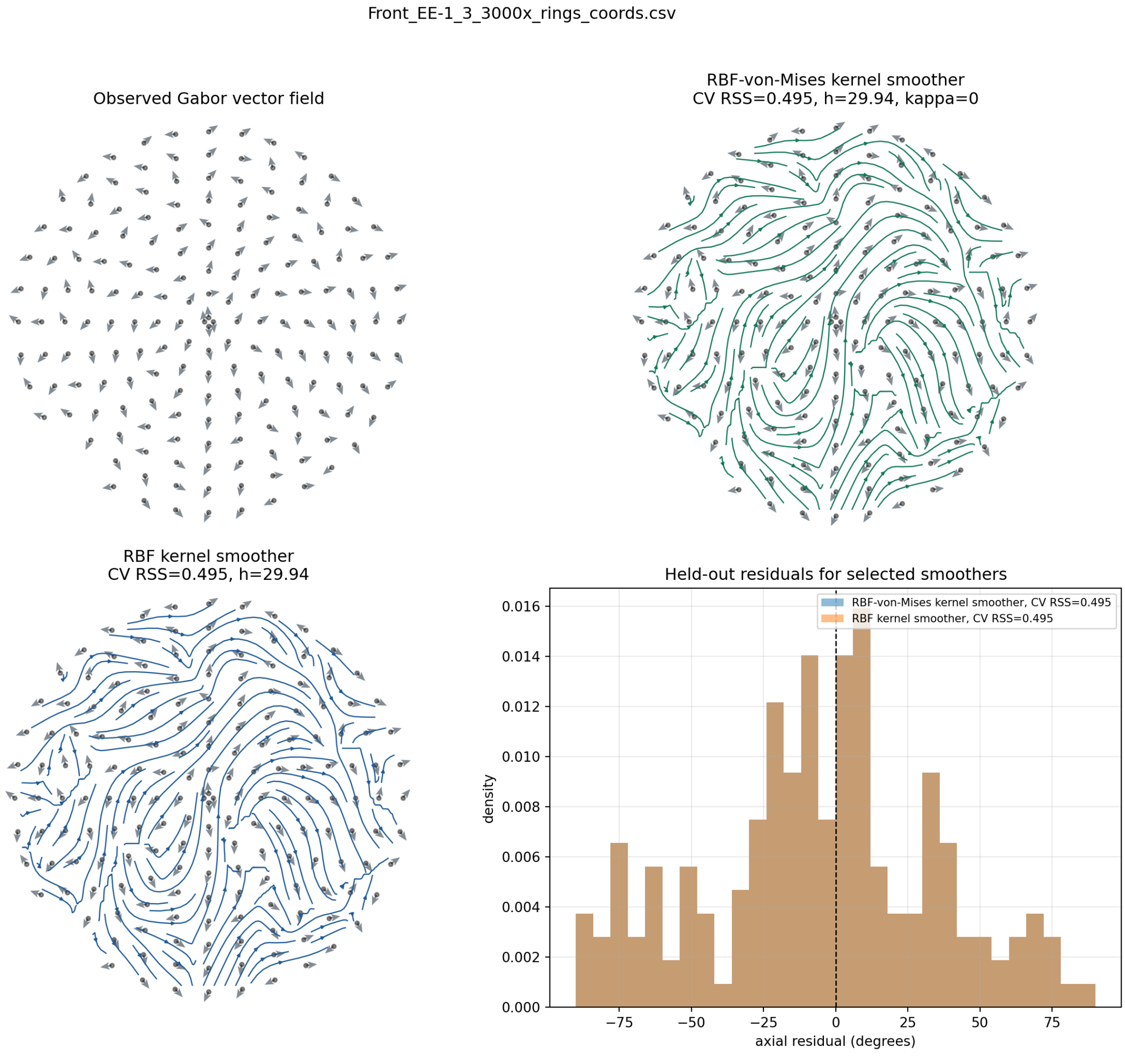

Figure 5: Kernel-smoother streamline overlays for `Front_EE-1_3_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.

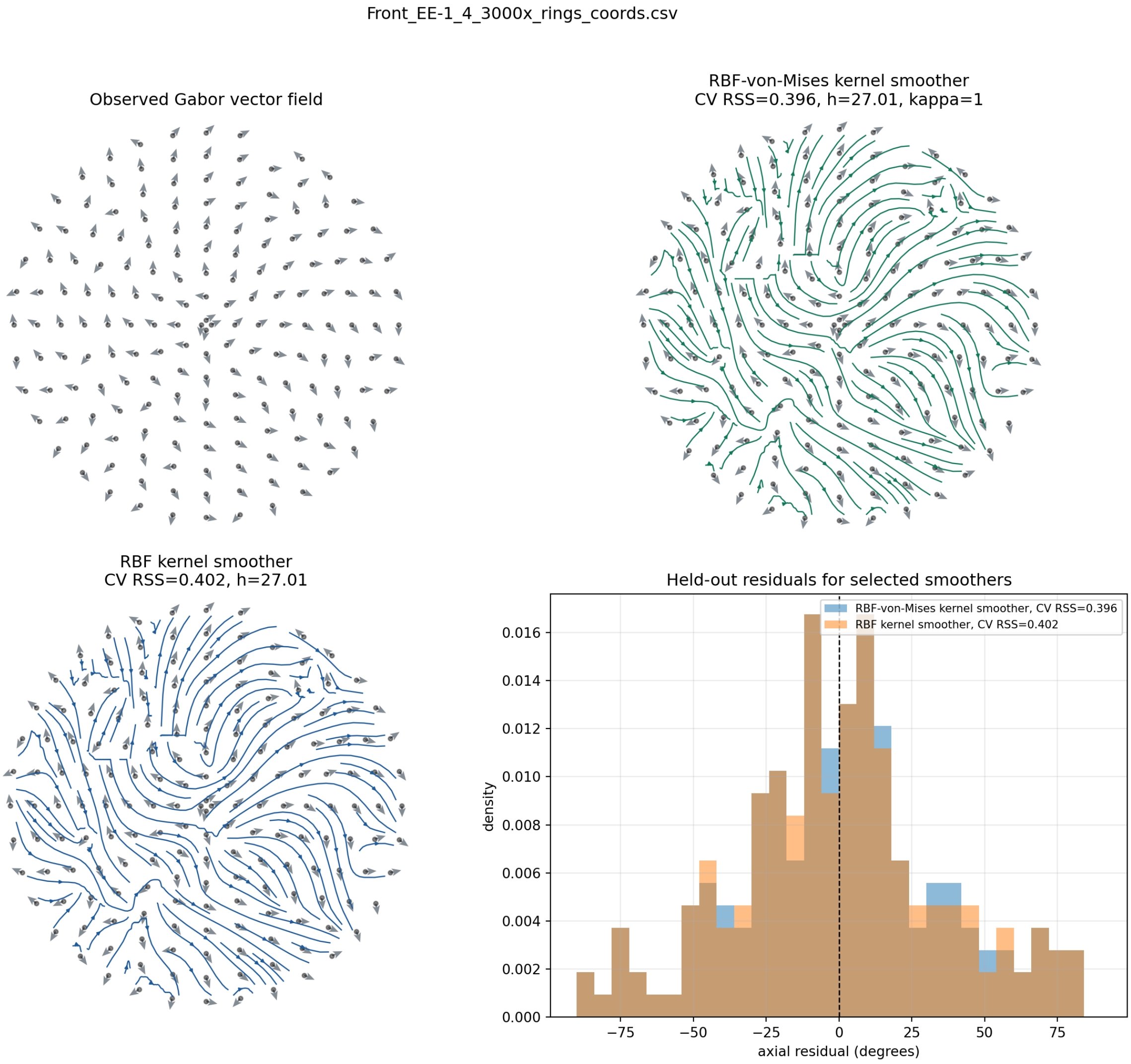

Figure 6: Kernel-smoother streamline overlays for `Front_EE-1_4_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.

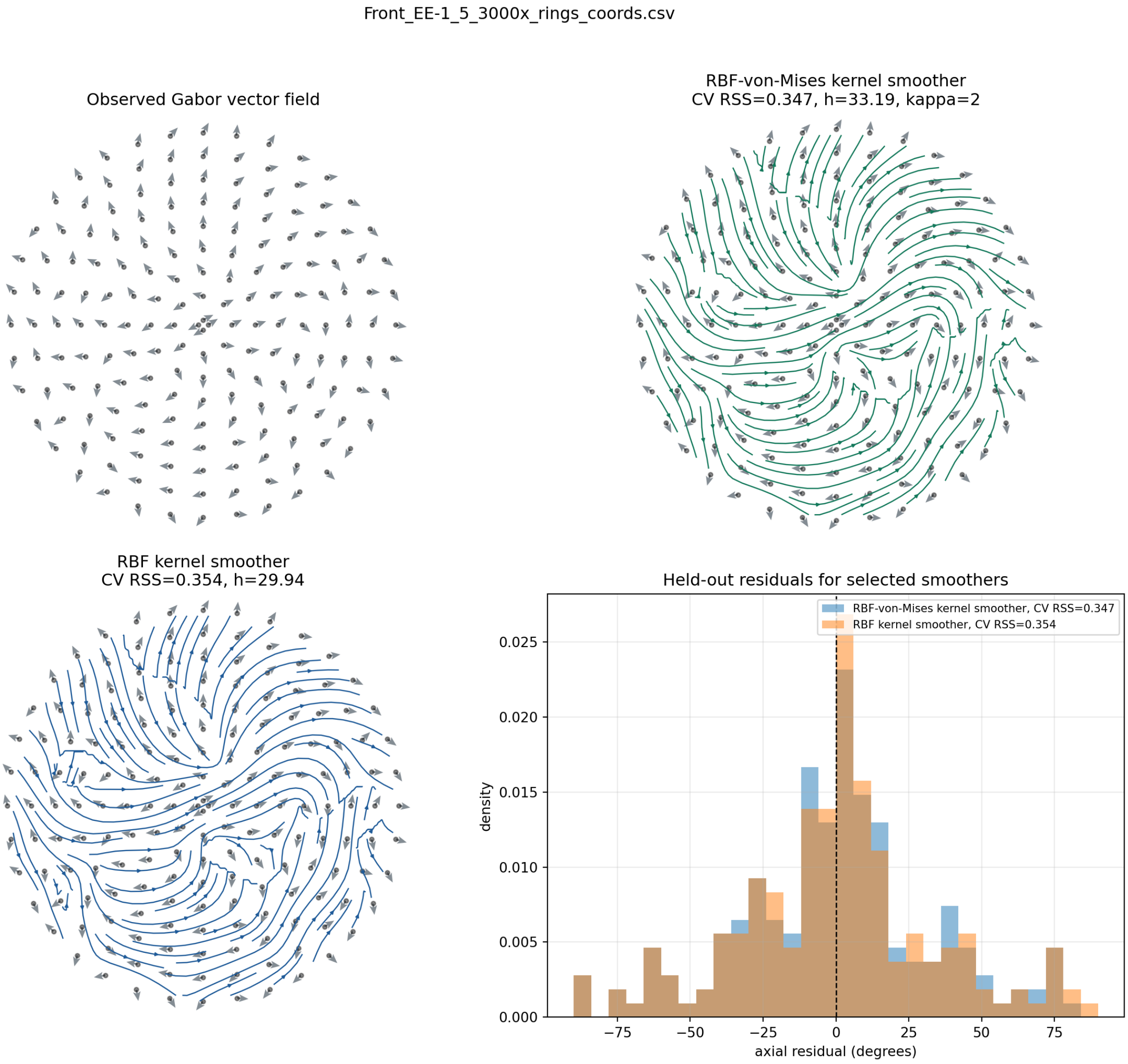

Figure 7: Kernel-smoother streamline overlays for `Front_EE-1_5_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.

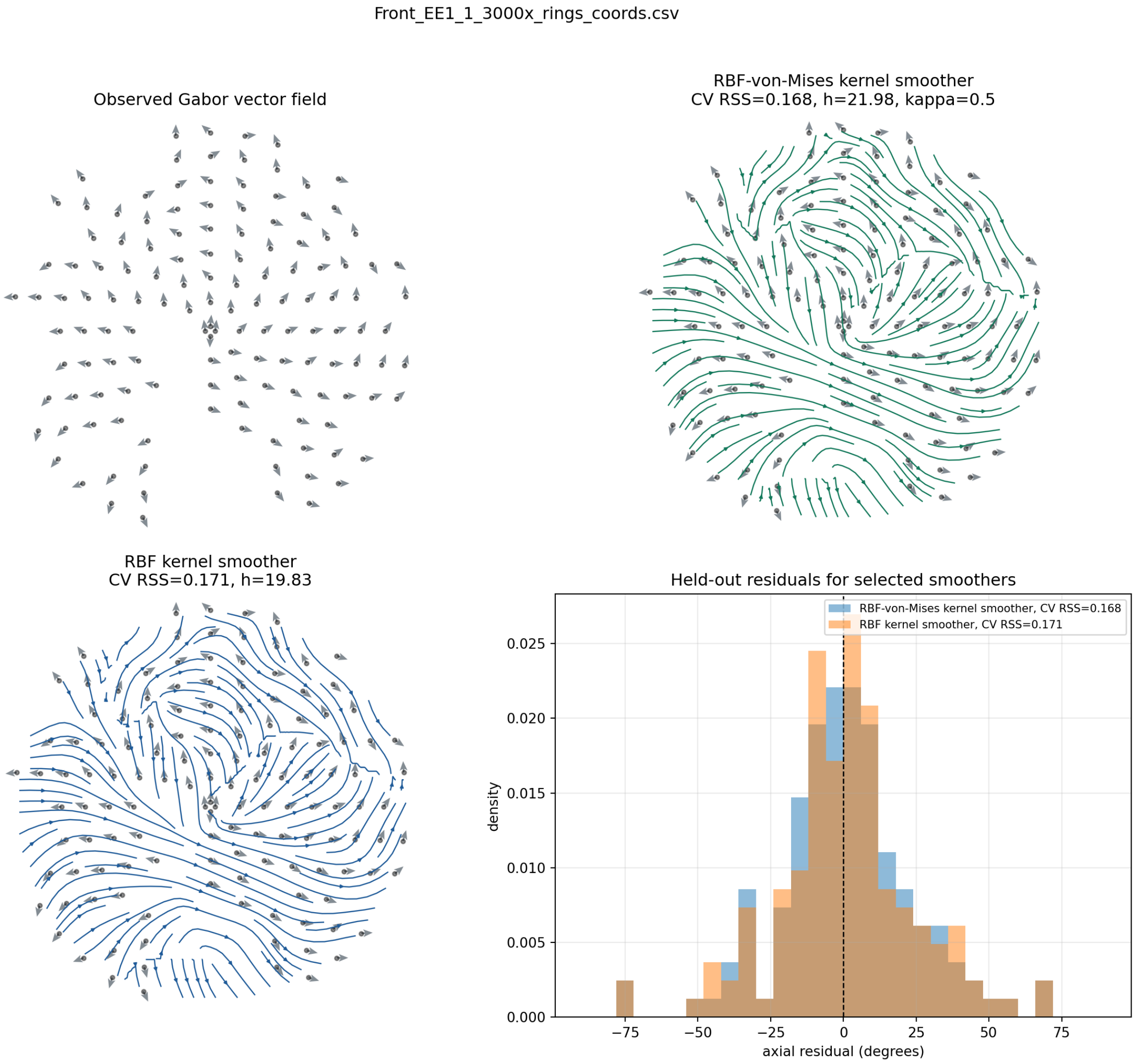

Figure 8: Kernel-smoother streamline overlays for `Front_EE1_1_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.

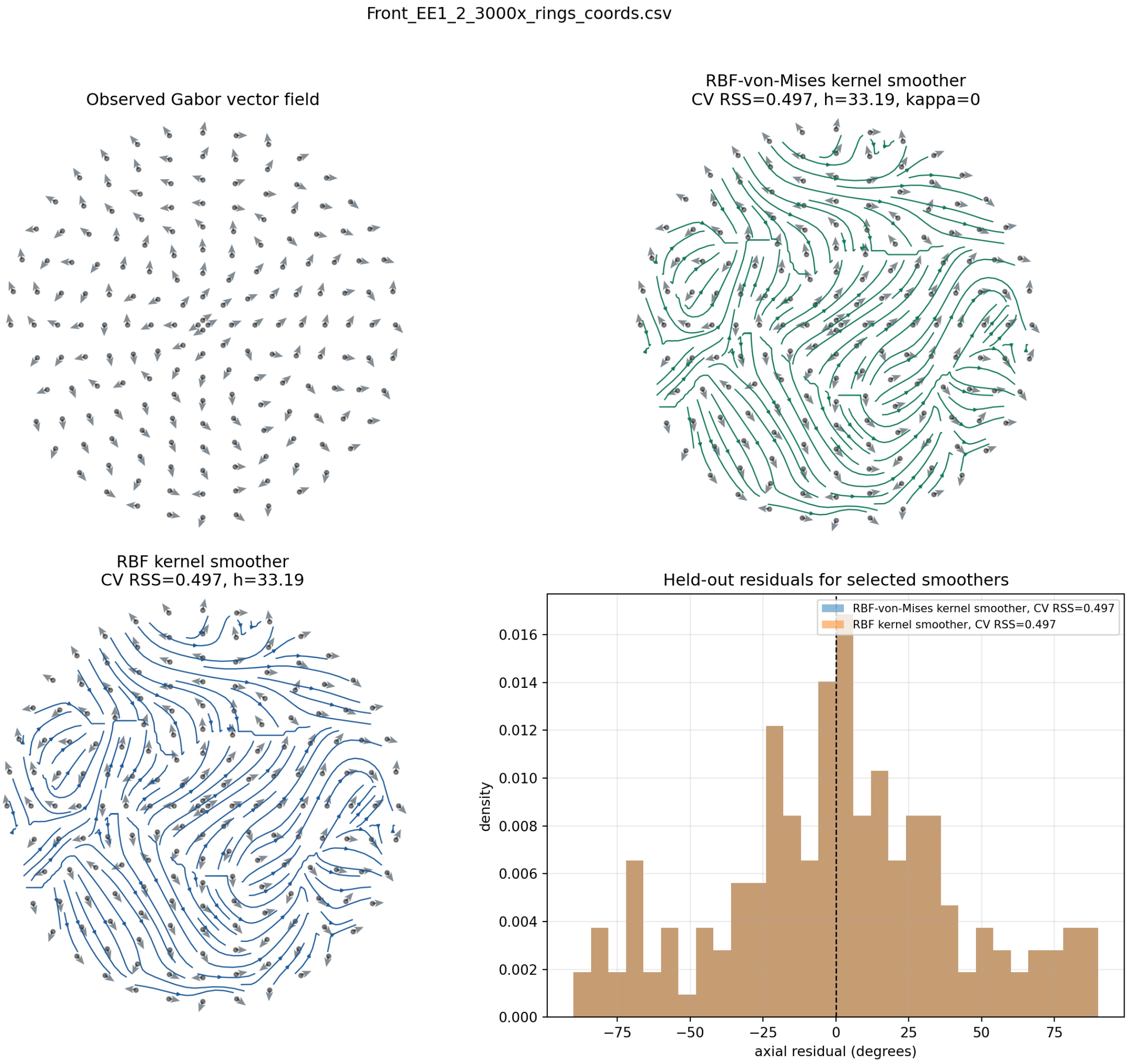

Figure 9: Kernel-smoother streamline overlays for `Front_EE1_2_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.

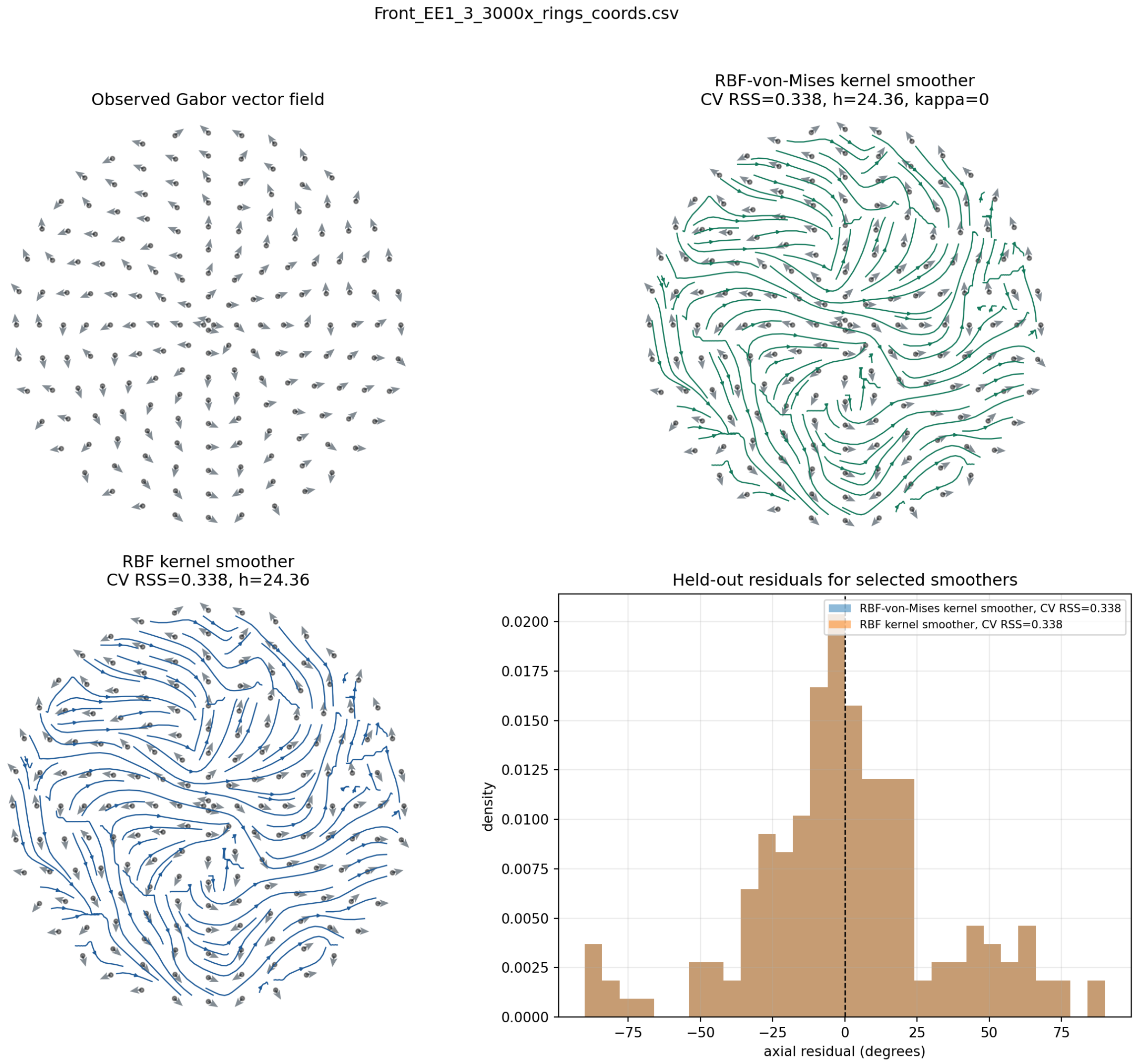

Figure 10: Kernel-smoother streamline overlays for `Front_EE1_3_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.

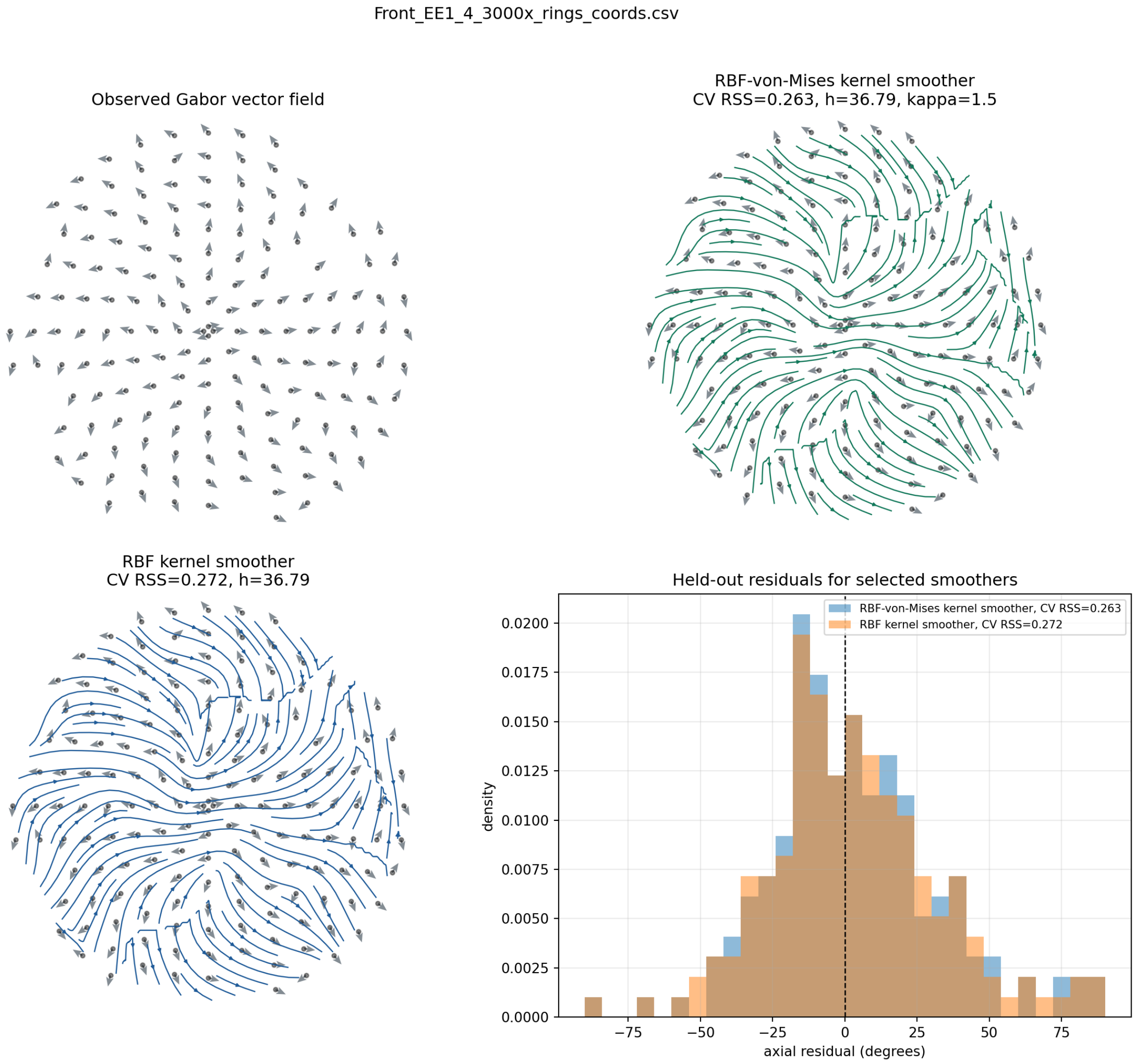

Figure 11: Kernel-smoother streamline overlays for `Front_EE1_4_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.

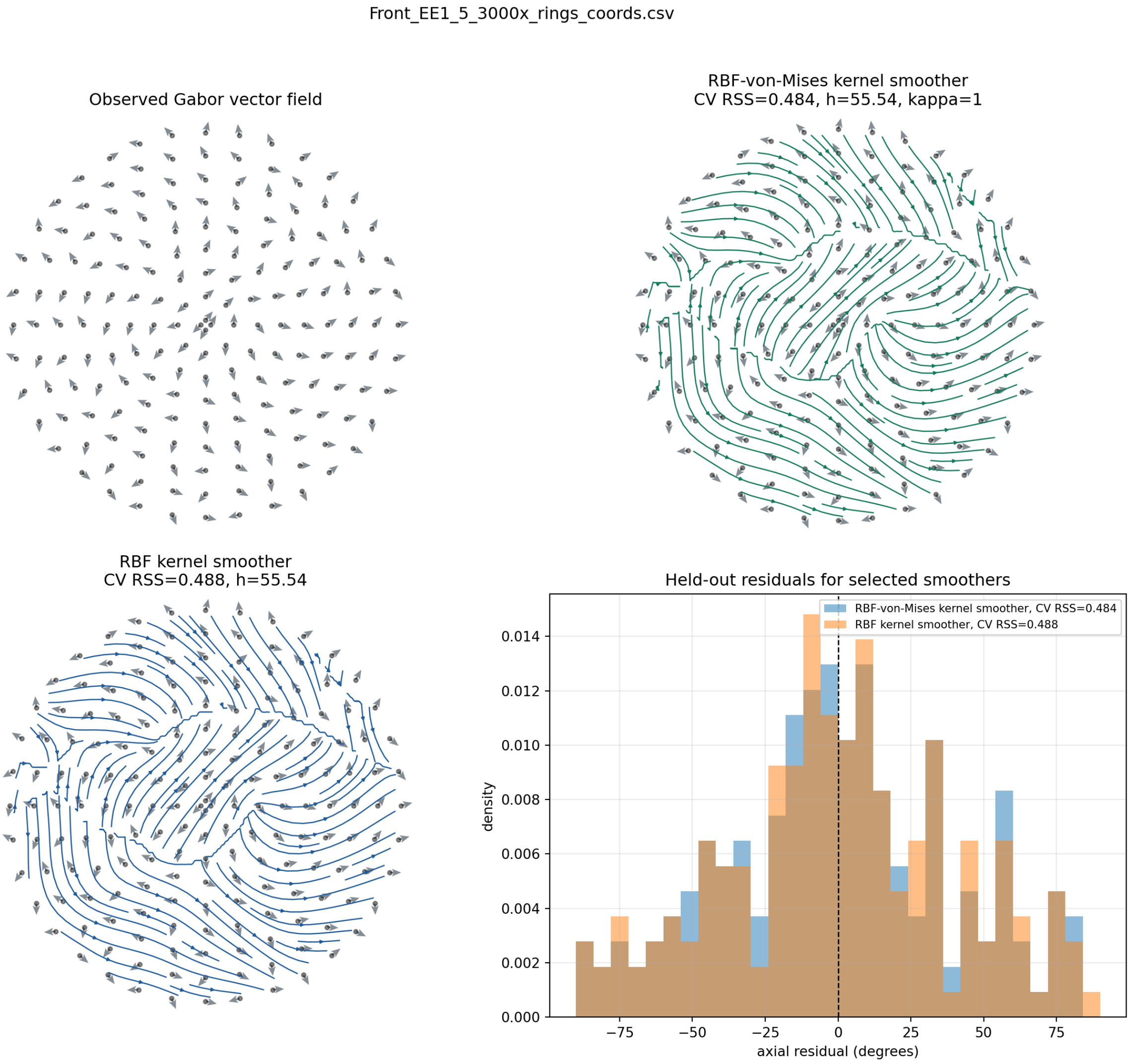

Figure 12: Kernel-smoother streamline overlays for `Front_EE1_5_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.